

Factor-Graph Aeromagnetic Compensation for Airborne Magnetic-Anomaly Navigation

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Abstract— Airborne magnetic-anomaly navigation bounds inertial drift in GNSS-denied flight by matching a scalar magnetometer against a magnetic-anomaly map. Its central obstacle is aeromagnetic interference: the aircraft’s own field must be compensated online when no calibration flight is available (the cold-start regime). A recent approach augments an extended Kalman filter (EKF) with online Tolles–Lawson (TL) coefficients and a residual neural network. We show that a factor-graph formulation reaches the same accuracy without a neural network by carrying the TL coefficients as graph variables of a fixed-lag sliding-window smoother, so the compensation adapts along the flight. Across four long lines from two flights and two maps, spanning free-flight navigation and survey missions, the smoother is the best of three cold-start estimators on every full-length line (8 of 8 cases): it stays bounded where a plain online-TL EKF diverges, and it beats a reimplemented EKF+TL+NN, which the neural network keeps bounded, without using a neural network. A simulation Monte-Carlo confirms that the reported covariance is consistent and never over-confident. We also give an exact linear condition for the joint observability of compensation and navigation, from which an exogeneity design rule for the compensation features follows.

Index Terms— Magnetic-anomaly navigation, aeromagnetic compensation, Tolles–Lawson, factor-graph optimization, fixed-lag smoothing, GNSS-denied navigation.

Nomenclature

$\mathbf{x}_t$ INS error state at epoch t (position, velocity, attitude, baro-inertial pair, IMU biases, FOGM), $\mathbf{x}_t \in \mathbb{R}^{18}$.

β_t Online Tolles–Lawson compensation coefficients.
 χ_t Joint state $(\mathbf{x}_t, \beta_t)$.
 $\Phi_t, \mathbf{Q}_t$ Pinson state-transition matrix and process covariance.
 z_t, η_t Scalar magnetometer measurement and its noise, variance R .
 $h(\cdot)$ Anomaly-map interpolation (with IGRF core) as a function of position.
 $\mathbf{g}_t$ Map gradient $\partial h / \partial \mathbf{p}$ at epoch t .
 $\mathbf{A}_t$ Tolles–Lawson basis row from the three-axis fluxgate.
 c_t Aircraft magnetic interference, $c_t = \mathbf{A}_t^\top \beta_t$.
 L_w, L_o Sliding-window length and overlap (look-ahead).
 ρ, δ Robust kernel and its Huber threshold.
 $\mathbf{G}, \Psi$ Stacked map-gradient and TL-regressor matrices over a window.

I. Introduction

INERTIAL navigation systems (INS) are the backbone of airborne navigation, but their position error grows without bound as accelerometer and gyroscope errors are integrated over time, so an INS must be aided by an absolute position reference [1], [2]. The satellite-based reference of choice, GNSS, is unavailable or untrustworthy in the contested and signal-degraded environments that increasingly define aerospace and defense operations. This has renewed interest in passive, self-contained references that draw on natural geophysical signatures of the Earth. Among these, the crustal magnetic-anomaly field is attractive: it is globally available, temporally stable over the mission horizon, difficult to deny or spoof, and observable with a lightweight scalar magnetometer [3], [4]. The same idea underlies terrain-referenced and gravity-referenced navigation, where an onboard measurement is matched against a stored geophysical map to bound inertial drift [5], [6], [7]; magnetic-anomaly navigation is distinguished by the ubiquity and fine spatial structure of the crustal field and by the maturity of airborne magnetic surveying. These properties make it a candidate absolute reference for long-endurance aircraft, uncrewed platforms, and munitions operating where GNSS cannot be trusted, provided the aircraft’s own magnetic signature can be removed in flight.

Magnetic-anomaly navigation (MagNav) estimates aircraft position by matching the measured scalar magnetic field against a surveyed anomaly map, yielding an absolute fix that bounds inertial drift [4]. Its central obstacle is not the map but the aircraft itself: the platform’s own permanent and induced magnetization and its eddy-current fields perturb the scalar measurement by hundreds of nanotesla, comparable to or larger than the anomaly signal being matched, so the aircraft field must be compensated before map matching can recover position. This aircraft-field problem is intrinsic to MagNav and independent of any particular airframe; it is most

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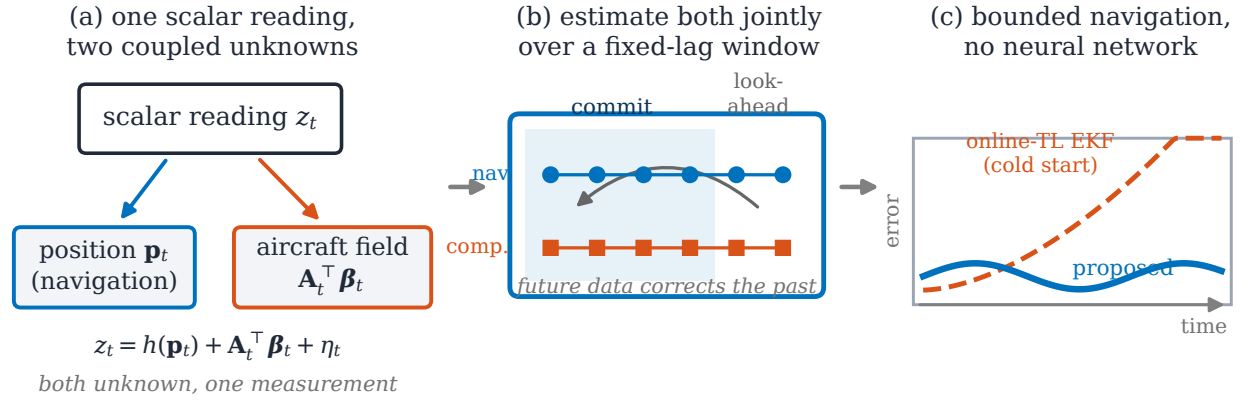


Fig. 1. Concept of the study. (a) A single scalar magnetometer reading mixes the anomaly map signal, which carries the aircraft position $\mathbf{p}_t$, with the aircraft field, modelled by the time-varying Tolles–Lawson term $\mathbf{A}_t^\top \boldsymbol{\beta}_t$; position and compensation are therefore coupled through the one measurement and must be estimated together. (b) The proposed estimator represents both as variables of a factor graph and solves them jointly over a fixed-lag sliding window, so that measurements after a given epoch still correct its navigation and compensation before the estimate is committed. (c) The result is bounded navigation at cold start, using no neural network, where a plain online-TL causal filter can diverge.

acute for magnetometers mounted inside the cabin, close to avionics and current-carrying structure, rather than on a wingtip stinger boom that holds the sensor away from the platform field.

Aeromagnetic compensation is classically handled by the Tolles–Lawson (TL) model [8], which regresses the interference onto sixteen attitude-dependent terms (permanent, induced, and eddy-current) formed from a three-axis fluxgate magnetometer. The coefficients are conventionally identified on a dedicated high-altitude calibration flight of prescribed maneuvers, where the anomaly signal is negligible and the interference dominates. Many operational scenarios, however, admit no such calibration flight: the coefficients drift with temperature and configuration, and a survey may begin with no prior calibration at all. In this cold-start regime the compensation coefficients must be learned online and jointly with navigation, from the same measurements used to fix position, a coupling that makes both estimation and its analysis substantially harder than the calibrated case.

A public benchmark has made these difficulties concrete and measurable: the Signal Enhancement for Magnetic Navigation (SGL 2020) challenge dataset [9], comprising flight-grade INS, multiple onboard magnetometers (both wingtip-stinger and cabin-mounted) and high-resolution anomaly maps, which has catalyzed most recent work on the problem. A second difficulty, prominent precisely on its cabin magnetometers, is that the linear TL basis does not span the true interference: platform currents and other non-rigid, non-attitude sources contribute a large fraction of the field that no set of Tolles–Lawson coefficients can represent [10]. This has motivated a line of learning-augmented compensation on this dataset: neural-network heading-error and denoising models [11], [10], and the online EKF+TL+NN whose residual network mops up exactly this un-modeled remainder [12]. The scalar optically-pumped magnetometers used for the cabin channels add their own orientation-dependent errors

(heading error and equatorial dead zones [13], [14]) on top of the platform field. How to absorb the non-TL remainder without a neural network is one of the questions this paper addresses.

Recursive Bayesian filtering is the standard machinery for map-matching navigation, from the linear Kalman filter [15] to nonlinear variants. Geophysical map matching has a long lineage: terrain-contour matching for cruise missiles [5] and nonlinear Kalman filtering for terrain-aided navigation [6] established the paradigm, and grid- and particle-based Bayesian estimators later became standard where the measurement-to-position map is strongly nonlinear or multimodal [16], [17], [18], [7]. For MagNav specifically, Canciani and Raquet combined a calibrated scalar magnetometer with an extended Kalman filter (EKF) and characterized the achievable accuracy on flight data [3], [4], later demonstrating airborne magnetic navigation with online calibration [19]. Beyond the EKF, unscented filtering [20], invariant formulations that respect the state manifold [21], and mixture/particle representations [22] improve consistency under nonlinearity. However, these estimators are causal: they commit each state online with no access to future measurements, and the position-centric ones carry no compensation state of their own. When a large uncompensated cold-start residual enters, a causal filter has no mechanism to attribute the mismatch to the aircraft field rather than to position, and the estimate can diverge.

A second line of work makes the compensation itself adaptive. Classical Tolles–Lawson (TL) identification is a batch regression on a calibration flight [8], [23], [24]; to remove that requirement the TL regression can be embedded in the estimator and its coefficients tracked online as additional states. Nonlinear extensions replace or augment the linear basis with learned components: neural-network compensators date back three decades [11], model-stability and robustness of the regression have been studied in gradiometry [25], and machine-learning-

enhanced calibration has recently been applied to the SGL benchmark [10]. Most relevant here, a residual neural network (NN) has been added on top of an online TL model, carried inside an EKF and trained online, to capture interference the linear basis misses [12]; this reaches good cold-start accuracy on the SGL 2020 cabin magnetometers and is, to our knowledge, the strongest reported cold-start result on that benchmark. Nevertheless, the network requires a carefully engineered online-training design to remain stable, adds opaque parameters whose behavior is hard to certify, and, being embedded in an EKF, inherits the same causal, commit-online limitation as the filters above.

A third body of work, largely from robotics, replaces filtering with factor-graph optimization: the estimation problem is a graph of variables and probabilistic constraints [26] whose maximum-a-posteriori (MAP) solution is found by sparse nonlinear least squares [27], [28], the same structure underlying bundle adjustment and graph SLAM [29], [30], [31]. Incremental and square-root smoothers (square-root SAM [32], iSAM [33], iSAM2 [34], and exactly-sparse delayed-state filters [35]) make the solution efficient enough for online use, and fixed-lag sliding-window optimization [36], [37] bounds its cost. On-manifold inertial preintegration [38], factor-graph inertial fusion [39], and modern visual-inertial estimators [40] have brought the approach firmly into navigation. Despite these advances, factor-graph estimation in navigation has been developed almost entirely for visual-inertial odometry and simultaneous localization and mapping, where the estimated quantities are poses and landmarks; it has not been brought to bear on joint online aeromagnetic compensation and map matching, and the observability of estimating a compensation model jointly with position from map matching has not been characterized.

Taken together, the three lines leave a specific gap. The cold-start MagNav problem is fundamentally a joint estimation of position and a time-varying compensation model from a single scalar stream, in which the two unknowns are confounded through the very measurement that must resolve them. Causal filters (cluster A) must commit each state before the maneuvers that would disambiguate the compensation have occurred, and so diverge when the initial residual is large; learned online compensation (cluster B) repairs that residual but keeps the causal, commit-online architecture and adds parameters that are difficult to certify; and factor-graph smoothing (cluster C), which can defer commitment and estimate structured nuisance variables jointly, has not been applied to this problem, nor has anyone characterized when the joint problem is even observable. What is missing is an estimator that (i) defers commitment so that late-arriving calibration information corrects early navigation states, (ii) represents the compensation (and the physical sensor errors of the scalar magnetometer) as first-class, time-varying variables rather than a black box, and (iii) comes with an explicit condition for when

position and compensation can be separated at all. This paper supplies all three.

In this work we cast online Tolles–Lawson compensation as a set of time-varying variables in a factor graph and solve the joint compensation-and-navigation problem as a fixed-lag sliding-window smoother (Fig. 1). The per-window relinearization supplies the time-varying compensation that the competing method obtains from an online network, and deferring commitment over the window removes the causal filter’s failure mode. We are explicit about scope: for the linear–Gaussian INS error model the batch MAP estimate on the factor-graph chain equals the Rauch–Tung–Striebel (RTS) smoother [41], so smoothing per se is not the contribution; the joint windowed estimation of a time-varying compensation is. The paper makes the following contributions:

- 1) **Neural-network-free online compensation.** We formulate online Tolles–Lawson compensation as a chain of time-varying variables in a fixed-lag sliding-window factor graph and solve navigation and compensation jointly. The per-window relinearization supplies the same time-varying compensation that the competing method draws from an online network, while deferring commitment over the window removes the causal filter’s cold-start failure mode. The result is an estimator that, on the uncompensated cabin magnetometers, matches or beats the online EKF+TL+NN of [12] without any learned component (Section III, Table III). We are deliberate about what is and is not claimed: for the linear–Gaussian error model the batch estimate equals the RTS smoother, so smoothing per se is not the contribution: the joint, windowed estimation of a time-varying compensation together with the robustness it buys at cold start is.
- 2) **Exact joint-observability condition.** We prove a necessary and sufficient linear-algebraic condition under which the aircraft position and the compensation coefficients are jointly observable from the scalar magnetometer, separating three distinct failure modes (an unexcited trajectory, a rank-deficient compensation basis, and the subtle one, a compensation subspace that is confounded with the anomaly gradient). From it we derive a concrete exogeneity design rule for choosing compensation features that a filter tuned by trial and error cannot see (Section IV).
- 3) **Reimplemented baseline and multi-line breadth study.** We reimplement, rather than merely cite, the online EKF+TL+NN result of [12] and place it head-to-head with the proposed smoother and a weak online-TL EKF on eight cold-start cases spanning four long lines, two flights, and two anomaly maps. The study isolates exactly what the neural network buys (it keeps the causal filter bounded where online TL alone diverges) and what it does not (on every full-length navigation line

the NN-free smoother is the most accurate of the three), giving a like-for-like comparison that the literature has lacked (Section VI).

- 4) **Physical sensor-error factors.** We add physically motivated optically-pumped-magnetometer error terms: a harmonic heading error carried as extra graph variables, and a dead-zone attenuation handled as a heteroscedastic reweighting of the scalar measurement. A controlled cumulative ablation then shows which of them are load-bearing. These are model terms that a position-only estimator cannot represent, and they let the graph absorb structured sensor error that would otherwise leak into the position estimate (Section III, Table VI).

The remainder of this paper is organized as follows. Section II states the INS error model, the measurement, and the batch MAP problem. Section III develops the sliding-window smoother and the compensation and sensor-error factors. Section IV establishes the observability condition. Section V describes the reimplemented EKF+TL+NN baseline, and Section VI reports and interprets the experiments. Section VII concludes and states the limitations.

II. Problem Formulation

We first fix notation and state the estimation problem. Throughout, boldface lowercase denotes vectors and boldface uppercase matrices; $\mathcal{N}(\cdot, \cdot)$ is the Gaussian density; and $\|\mathbf{v}\|_M^2 = \mathbf{v}^\top \mathbf{M} \mathbf{v}$ is the squared Mahalanobis norm.

A. INS error state and dynamics

We adopt the standard Pinson error model of a strap-down INS [2], [1], [42]. The error state

$$\mathbf{x} = [\delta \mathbf{p}^\top \delta \mathbf{v}^\top \psi^\top h_a \hat{a} \mathbf{b}_a^\top \mathbf{b}_g^\top S]^\top \in \mathbb{R}^{18} \quad (1)$$

stacks position error $\delta \mathbf{p}$, velocity error $\delta \mathbf{v}$, attitude error ψ (three components each), the baro-inertial altitude-aiding pair $(h_a, \hat{a})$ that damps the vertically unstable channel, accelerometer and gyroscope biases $\mathbf{b}_a, \mathbf{b}_g$, and a scalar first-order Gauss–Markov (FOGM) state S that absorbs residual magnetic disturbance and map error. The discretized, linearized dynamics are

$$\mathbf{x}_{t+1} = \Phi_t \mathbf{x}_t + \mathbf{w}_t, \quad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_t), \quad (2)$$

where Φ_t is the Pinson state-transition matrix (Appendix A) and $\mathbf{Q}_t$ the process covariance; the FOGM block of Φ_t is $e^{-\Delta t/\tau}$ with correlation time τ .

B. Scalar magnetometer measurement

The scalar magnetometer reads

$$z_t = h(\mathbf{p}_t) + c_t + \eta_t, \quad \eta_t \sim \mathcal{N}(0, R), \quad (3)$$

where $h(\cdot)$ interpolates the crustal anomaly map (with the IGRF core field restored) at the aircraft position $\mathbf{p}_t =$

$\bar{\mathbf{p}}_t + \delta \mathbf{p}_t$ and c_t is the aircraft interference. Linearizing about the nominal position,

$$z_t \approx h(\bar{\mathbf{p}}_t) + \mathbf{g}_t^\top \delta \mathbf{p}_t + c_t + \eta_t, \quad \mathbf{g}_t = \left. \frac{\partial h}{\partial \mathbf{p}} \right|_{\bar{\mathbf{p}}_t}, \quad (4)$$

so the map gradient $\mathbf{g}_t$ is the sole source of horizontal position information: where $\mathbf{g}_t \approx \mathbf{0}$ (a locally flat map) position is momentarily unobservable, and everywhere the interference c_t competes with the $\mathbf{g}_t^\top \delta \mathbf{p}_t$ term for the same measurement. The map h is stored as a gridded anomaly product upward-continued to the survey altitude; we evaluate it and its gradient by bicubic interpolation, which yields the continuous $\mathbf{g}_t$ required to linearize (4), and we restore the IGRF core field so that z_t and h are on a common absolute scale. The gradient magnitude $\|\mathbf{g}_t\|$ varies by more than an order of magnitude along a line, which is why position accuracy is not uniform in time even for a perfect compensator, a point the error traces of Section VI make visible.

C. Tolles–Lawson interference model

Classical aeromagnetic compensation writes the interference as a linear combination of attitude-dependent basis terms [8], [23], [24],

$$c_t = \mathbf{A}_t^\top \boldsymbol{\beta}_t, \quad (5)$$

where $\mathbf{A}_t \in \mathbb{R}^m$ is the TL basis row of (5) built from the three-axis fluxgate; its classical rank deficiency is a gauge freedom (Remark 2). With direction cosines $\hat{\mathbf{b}}_t = [u \ v \ w]^\top$ of the Earth field in body axes and total field magnitude B_t , the permanent (3), induced (6), and eddy-current (9) groups are

$$\mathbf{A}_t = \left[\underbrace{u, v, w}_{\text{perm.}}, \underbrace{B_t u^2, \dots, B_t w^2}_{\text{induced}}, \underbrace{B_t u \dot{u}, \dots, B_t w \dot{w}}_{\text{eddy}} \right]^\top, \quad (6)$$

optionally augmented by a bias term, giving $m \leq 19$ coefficients β_t . In the calibrated setting $\boldsymbol{\beta}$ is fit once and frozen; in the cold-start setting it is unknown at takeoff and drifts slowly, so we treat it as a time-varying state with a random-walk prior $\boldsymbol{\beta}_{t+1} = \boldsymbol{\beta}_t + \mathbf{w}_t^\beta$, $\mathbf{w}_t^\beta \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}^\beta)$.

D. Batch maximum-a-posteriori estimate

Collecting the navigation and compensation variables into $\boldsymbol{\chi}_t = (\mathbf{x}_t, \boldsymbol{\beta}_t)$ over a window of N epochs, the MAP estimate minimizes

$$J(\boldsymbol{\chi}) = \|\boldsymbol{\chi}_1 - \boldsymbol{\chi}_0\|_{\mathbf{P}_0^{-1}}^2 + \sum_{t=1}^{N-1} \|\mathbf{x}_{t+1} - \Phi_t \mathbf{x}_t\|_{\mathbf{Q}_t^{-1}}^2 + \sum_{t=1}^{N-1} \|\boldsymbol{\beta}_{t+1} - \boldsymbol{\beta}_t\|_{(\mathbf{Q}^\beta)^{-1}}^2 + \sum_{t=1}^N \rho \left(\frac{r_t(\boldsymbol{\chi}_t)}{\sqrt{R}} \right), \quad (7)$$

with measurement residual $r_t(\boldsymbol{\chi}_t) = z_t - h(\mathbf{p}_t) - \mathbf{A}_t^\top \boldsymbol{\beta}_t - S_t$ and ρ a possibly robust kernel; the prior term anchors

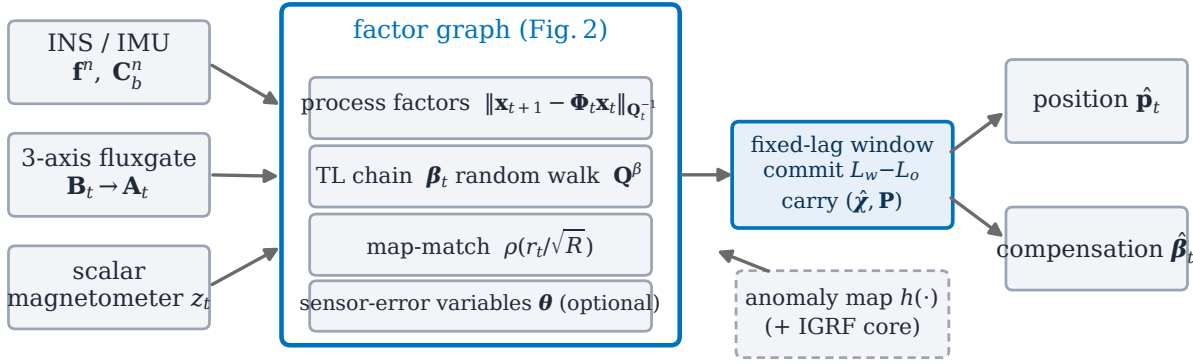


Fig. 2. Proposed estimator. The INS error state, the three-axis fluxgate (Tolles–Lawson basis), and the scalar magnetometer feed a factor graph whose variables are the navigation error, the online TL coefficients, and optional sensor-error states. A fixed-lag window solves the graph by robust Gauss–Newton/QR and outputs position and compensation; the anomaly map enters through the map-match factor.

both blocks of $\chi_1 = (\mathbf{x}_1, \beta_1)$, at cold start with β -block $\mathbf{P}_0^\beta$ (Table I). We make the standard modeling assumptions of aided inertial navigation [1], [22].

Assumption 1:

The process and measurement noises $\mathbf{w}_t, \mathbf{w}_t^\beta, \eta_t$ are zero-mean, white, mutually independent, and Gaussian; the linearizations (2) and (4) are valid over a window; and the prior $(\hat{\chi}_0, \mathbf{P}_0)$ is Gaussian.

Assumption 1 is the same one under which the EKF and RTS smoother are derived, so any performance difference between the estimators reflects their structure, not their noise models. Under it the batch estimate coincides with the classical smoother.

Lemma 1:

Under Assumption 1, with $\rho(u) = \frac{1}{2}u^2$ and β held fixed, the minimizer of (7) equals the Rauch–Tung–Striebel (RTS) smoother estimate over the window.

Proof:

With ρ quadratic and β fixed, (7) is a positive-definite quadratic in the Gaussian chain $\{\mathbf{x}_t\}$, so its minimizer is the posterior mean $\mathbb{E}[\mathbf{x}_{1:N} | z_{1:N}]$. For a linear–Gaussian state-space model that conditional mean is exactly what the forward Kalman recursion followed by the backward RTS recursion computes [41]; the two therefore coincide. ■

Remark 1:

Lemma 1 is why we do not claim smoothing itself as a contribution: for a static, known compensation the batch graph and a filter–smoother pass are equivalent. The contribution is what the graph makes easy that a filter does not: jointly estimating the time-varying β , adding the non-Gaussian robust kernel of (10), and appending the sensor-error variables of Section III, together with the fixed-lag windowing that keeps it causal-latency bounded.

III. Method

The proposed estimator (Fig. 2) represents navigation, online compensation, and optional sensor-error states as variables of a single factor graph, and solves it over a fixed-lag sliding window by robust Gauss–Newton. This section develops the factors, the windowing, the robust solver, and the sensor-error terms.

A. Online TL coefficients as graph variables

We append the compensation coefficients β to the state, so the measurement factor of (3) becomes, in residual form,

$$r_t(\chi_t) = z_t - h(\mathbf{p}_t) - \mathbf{A}_t^\top \beta_t - S_t, \quad (8)$$

with Gaussian information R^{-1} , and its Jacobian row is

$$\mathbf{H}_t = \begin{bmatrix} \underbrace{\mathbf{g}_t^\top}_{\delta \mathbf{p}} & \underbrace{\mathbf{0}}_{\delta \mathbf{v}, \psi, h_a, \hat{a}, \mathbf{b}} & \underbrace{\mathbf{A}_t^\top}_{\beta} & \underbrace{1}_S \end{bmatrix}. \quad (9)$$

The window posterior over the joint states $\chi_t = (\mathbf{x}_t, \beta_t)$ is the product of four factor families, drawn explicitly in Fig. 3:

- 1) a joint prior on χ_1 : $\mathbf{P}_0$ at cold start, or the smoothed mean and covariance carried from the previous window (Section III-B);
- 2) Pinson–FOGM process factors $\|\mathbf{x}_{t+1} - \Phi_t \mathbf{x}_t\|_{\mathbf{Q}_t^{-1}}^2$, parameterized by the INS mechanization (specific force $\mathbf{f}^n$, attitude $\mathbf{C}_b^n$) through (12);
- 3) compensation random-walk factors $\|\beta_{t+1} - \beta_t\|_{(\mathbf{Q}^\beta)^{-1}}^2$, which set how quickly the compensation may adapt: $\mathbf{Q}^\beta \rightarrow \mathbf{0}$ recovers a single static β ;
- 4) robust scalar-magnetometer factors $\rho(r_t/\sqrt{R})$ with the residual (8), each coupling $\delta \mathbf{p}_t$ (through the map gradient $\mathbf{g}_t$), the disturbance state S_t , and β_t (through the fluxgate regressor $\mathbf{A}_t$).

The INS mechanization, the fluxgate (through $\mathbf{A}_t$), and the anomaly map (through h and $\mathbf{g}_t$) enter as known factor parameters, not as estimated variables; the optional

sensor-error variables θ of Section III-F attach to every measurement factor when enabled. Two implementation details: in the online estimator only the horizontal components of $\mathbf{g}_t$ are used (the vertical channel is baro-damped and the map is evaluated at survey altitude), whereas the batch validation of Section VI-A uses the full three-component gradient; and the implementation orders the state vector as $[\text{Pinson-17}; \beta; S]$, a permutation of (1) that does not affect the estimate. Because consecutive coefficient blocks are chained rather than instantaneous, the compensation is identified from the whole trajectory, and because the batch solution uses future as well as past measurements, early navigation states benefit from calibration information that only becomes observable after later maneuvers, the property a causal filter lacks.

Absorbing the non-Tolles–Lawson remainder. On this dataset the cabin interference is dominated by platform currents and other sources the attitude-only TL basis $\mathbf{A}_t$ cannot span, and the online EKF+TL+NN of [12] devotes a neural network precisely to this remainder. The proposed graph carries no network; instead three of the factors above combine to absorb the remainder as a structured, robust regression problem. The random-walk factor (iii) makes β_t time-varying: writing the true interference as $\mathbf{A}_t^\top \beta^* + e_t$, a static fit ($\mathbf{Q}^\beta \rightarrow \mathbf{0}$) forces the entire current signature into the residual e_t , whereas a nonzero $\mathbf{Q}^\beta$ lets the estimated coefficients track the slowly drifting, maneuver-correlated projection of the currents onto $\mathbf{A}_t$, the low-frequency part a single calibration fit cannot recover. The disturbance state S_t in the residual (8), a first-order Gauss–Markov process folded into the Pinson–FOGM factor (ii), then carries the temporally correlated component of e_t that is orthogonal to $\mathbf{A}_t$; it is the linear–Gaussian, single-scalar analogue of the learned residual, cheaper and far less expressive but requiring no training data. What neither term explains, the heavy-tailed switching transients of onboard electronics, is handled not by fitting but by the robust kernel in factor (iv), which the reweighting of Section III-D down-weights so that such samples cannot pull the position estimate. The observability condition of Section IV is what keeps this well-posed: S_t and β_t may only absorb signal that is not collinear with the map gradient $\mathbf{g}_t$, so genuine map information is not silently explained away as interference. The trade it makes, a higher compensation residual than the network in exchange for navigation robustness, is quantified with the cold-start results in Section VI-C.

B. Fixed-lag sliding window

A single batch over a long line forces one static β , which underfits time-varying interference; a per-epoch filter, conversely, cannot use future data. We take the middle path and process the flight in overlapping windows of length L_w with overlap L_o , each a batch factor graph, in the manner of fixed-lag and incremental smoothing [34], [36], [37]. Window k spans epochs $[s_k, s_k + L_w]$; it is solved as a batch, commits only its leading stride of

Algorithm 1 Fixed-lag sliding-window smoother

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1: Input: measurements  $\{z_t\}$ , TL rows  $\{\mathbf{A}_t\}$ , map  $h$ ,
   window  $L_w$ , overlap  $L_o$ , prior  $(\hat{\chi}_0, \mathbf{P}_0)$ 
2:  $s \leftarrow 1$ 
3: while  $s \leq T$  do
4:   build factor graph on epochs  $[s, \min(s+L_w, T)]$ 
   with prior  $(\hat{\chi}_{s-1}, \mathbf{P}_{s-1})$ 
5:   repeat ▷ robust Gauss–Newton
6:     linearize residuals (8); form weights  $w_t$  from
        $\rho$  (11)
7:     solve the linearized chain by an iterated RTS
       pass [41] (equivalently, sparse QR / square-root
       SAM [32]; Sec. III-E)
8:     update  $\chi \leftarrow \chi + \Delta\chi$ 
9:   until converged
10:  commit epochs  $[s, s+L_w-L_o]$ ; recover  $(\hat{\chi}, \mathbf{P})$  at
     epoch  $s+L_w-L_o-1$ 
11:   $s \leftarrow s + (L_w - L_o)$ 
12: end while
13: Output: committed trajectory  $\{\hat{\mathbf{p}}_t, \hat{\beta}_t\}$ 

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$L_w - L_o$ epochs, and carries the full smoothed state and covariance of the last committed epoch forward as a Gaussian prior for window $k+1$ (Fig. 4). The trailing overlap L_o acts as bounded smoother look-ahead, so each committed estimate has seen L_o epochs of future data. Carrying the full covariance, not just the mean, keeps navigation continuous across window boundaries. One approximation deserves note: the carried prior is the smoothed state at the commit boundary, which has already seen the overlap measurements, and window $k+1$ processes those L_o epochs again, so the overlap information is counted twice. An exactly consistent scheme would carry the forward-filtered state or marginalize the overlap; we accept the bounded optimism (the overlap is a fixed fraction, $L_o/L_w = 0.3$, of each window) in exchange for the simpler implementation, and note that the reported DRMS is a comparison of means, unaffected by the covariance bookkeeping. Algorithm 1 summarizes the procedure; its per-window cost is constant in the line length, giving overall linear complexity.

C. Window design and initialization

Two parameters govern the window: the length L_w and the overlap L_o . L_w sets how much data each compensation estimate sees and thus the effective adaptation time; too short and β is under-determined, too long and it cannot track drift. L_o sets the look-ahead and hence the commit latency: a committed state has seen L_o epochs of future data, and the estimator lags real time by L_o . In the experiments we use $L_w = 5$ min and $L_o = 90$ s; the 2 min window in Table III shows that shortening L_w degrades the long-line result, consistent with under-determination. The first window is seeded with a diffuse prior on β (large $\mathbf{P}_0$ block) so the cold start is not biased toward

INS error state $\mathbf{x}_t = [\delta \mathbf{p} \ \delta \mathbf{v} \ \psi \ h_a \ \hat{\mathbf{a}} \ \mathbf{b}_a \ \mathbf{b}_g \ S]^T \in \mathbb{R}^{18}$

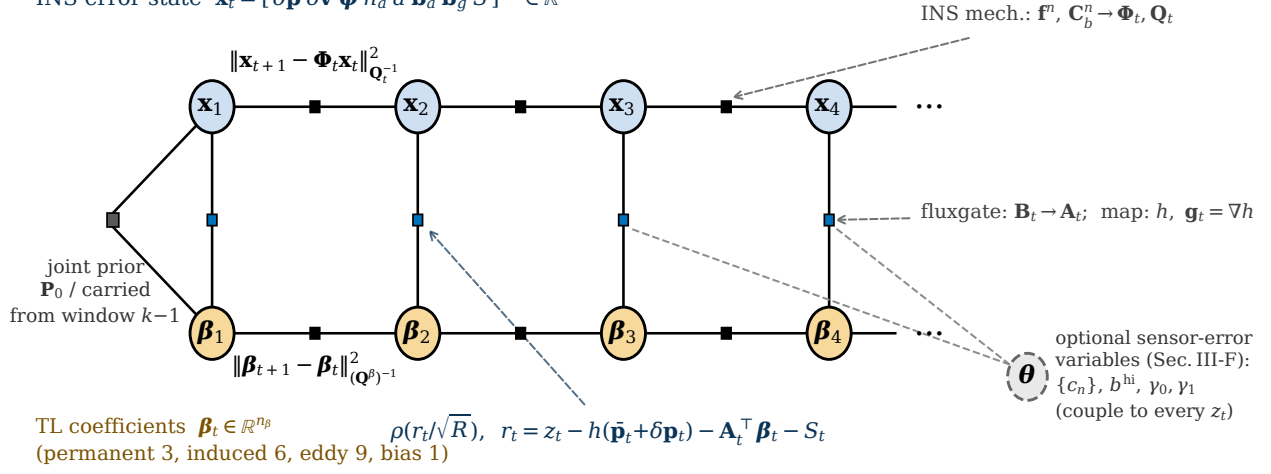


Fig. 3. Factor graph of one window, with every estimated variable and measurement shown. Top: the 18-state INS error chain $\mathbf{x}_t$ of (1), linked by Pinson-FOGM process factors driven by the INS mechanization. Bottom: the time-varying Tolles-Lawson coefficient chain β_t , linked by random-walk factors $\mathbf{Q}^\beta$. Each scalar-magnetometer factor (blue) applies the robust residual of (8), coupling $\delta \mathbf{p}_t$, S_t , and β_t ; the joint prior (left) is $\mathbf{P}_0$ at cold start or the covariance carried from the previous window. Dashed arrows mark known factor parameters (INS mechanization, fluxgate basis $\mathbf{A}_t$, map $h, \nabla h$); the dashed node θ holds the optional sensor-error variables of Section III-F.

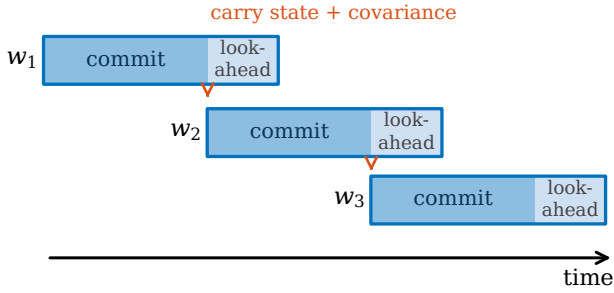


Fig. 4. Fixed-lag sliding window. Each window commits its leading stride and passes the full smoothed state and covariance forward as the prior for the next window; the trailing overlap provides bounded smoother look-ahead.

zero interference, and the FOGM disturbance state absorbs the initial mismatch while the coefficients converge; subsequent windows inherit the tight carried covariance and converge in a few Gauss-Newton iterations (typically 3–5; Sec. VI-F). This warm-start is why the per-window iteration count stays low despite the nonlinearity of the map term.

D. Robust kernel and reweighting

Map error, transient interference, and dead-zone measurements produce heavy-tailed residuals that a quadratic cost over-weights. We use a Huber kernel [43]

$$\rho(u) = \begin{cases} \frac{1}{2}u^2, & |u| \leq \delta, \\ \delta(|u| - \frac{1}{2}\delta), & |u| > \delta, \end{cases} \quad (10)$$

minimized by iteratively reweighted least squares (IRLS): each Gauss-Newton iteration reweights measurement factor t by

$$w_t = \min(1, \delta/|u_t|), \quad u_t = r_t/\sqrt{R}, \quad (11)$$

so gross outliers are down-weighted toward an absolute-value penalty while inliers keep the efficient quadratic cost. In the implementation the reweighting is applied from the second Gauss-Newton iteration onward (the first iteration is plain least squares) with w_t clamped to $[10^{-6}, 1]$, and $\delta = 1.345$ gives the standard 95% Gaussian efficiency. Alternatives such as switchable constraints [44] and dynamic covariance scaling [45] fit the same graph interface and could replace (10) unchanged.

E. Sparse solver

Each Gauss-Newton iteration solves a sparse linear system whose Jacobian has the block-bidiagonal structure of the two chains in Fig. 3 (in the static- β limit the coefficient blocks collapse to one shared column). We factor it by column-pivoted QR on the square-root information matrix (square-root SAM [32], [27]), which is numerically stable and exploits the chain sparsity; an iterated RTS pass [41] gives the identical estimate for the Gaussian case and we verify the two agree to within a metre on a real-data segment in Section VI (the check covers the navigation-only chain; the joint TL problem is solved by the iterated-RTS path).

F. Physical sensor-error factors

The graph representation admits error states that a position-only grid or point-mass estimator cannot carry. Beyond TL we optionally augment χ with three physically motivated scalar-magnetometer error variables, each appearing additively in the residual (8), plus one measurement re-weighting:

- 1) OPM heading error, modeled as the cosine-harmonic expansion of the optically-

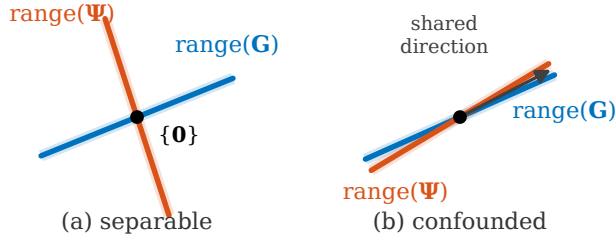


Fig. 5. Observability of joint compensation and navigation (Proposition 1). Left: when the map-gradient subspace $\text{range}(\mathbf{G})$ and the compensation subspace $\text{range}(\Psi)$ meet only at the origin, position and compensation are separable. Right: when they share a nonzero direction, a position shift can be masked by a compensation change and observability collapses.

pumped (scalar-atomic) magnetometer in the sensor-field angle $\psi_t = \angle(\hat{\mathbf{e}}, \mathbf{B}_t)$, $b_t^{\text{hdg}} = \sum_{n \in \{1,2,4\}} c_n \cos(n\psi_t)$ [13], [14], with coefficients $\{c_n\}$ as graph variables; the $n=1$ term is the vector light shift and $n=2$ the nonlinear Zeeman effect. Because ψ_t is known from attitude and IGRF, the term is linear in $\{c_n\}$ and, as in the Tolles–Lawson case [14], enters the graph as an ordinary linear regressor;

- 2) Hard-iron bias b^{hi} , a constant offset variable;
- 3) Linear drift $b_t^{\text{dr}} = \gamma_0 + \gamma_1 t$, capturing slow electronics drift;
- 4) Dead-zone weight: not a graph variable but a heteroscedastic measurement information $R_t^{-1} = R^{-1} \max(|\cos \psi_t|, \epsilon)^2$ ($\epsilon=0.05$) that follows the OPM signal-amplitude collapse near the equatorial dead zone (field $\perp$ optical axis) [13] and de-weights those samples (a de-weighting any estimator could apply; it is included for completeness of the sensor model).

The augmented residual is $r_t = z_t - h(\mathbf{p}_t) - \mathbf{A}_t^\top \beta_t - S_t - b_t^{\text{hdg}} - b^{\text{hi}} - b_t^{\text{dr}}$. Each term is exogenous to position (Section IV), so it can be identified without stealing map information; their observability is limited only by how much the flight excites the corresponding regressor (e.g. the heading harmonics need attitude variation).

IV. Observability of Joint Compensation and Navigation

Whether position can be recovered while the compensation is estimated jointly is a property of the measurement geometry. Over a window let $\mathbf{G}$ have rows $\mathbf{g}_t^\top$ (the map-gradient/position sensitivity) and let Ψ have rows $\mathbf{A}_t^\top$ (the TL regressor). To isolate the position–compensation coupling, restrict (3) to the $(\delta\mathbf{p}, \delta\beta)$ subspace, giving the window model $\mathbf{z} = \mathbf{G}\delta\mathbf{p} + \Psi\delta\beta + \boldsymbol{\eta}$ (Fig. 5). This model freezes $\delta\mathbf{p}$ and $\delta\beta$ over the window: it is a per-window surrogate for the time-varying system (2), (8) that isolates the position–compensation coupling which a full observability Gramian would mix with the INS dynamics.

Proposition 1:

A position perturbation $\delta\mathbf{p}$ is indistinguishable from a compensation adjustment, hence unobservable from map matching over the window, if and only if $\mathbf{G}\delta\mathbf{p} \in \text{range}(\Psi)$. Consequently the joint state $(\delta\mathbf{p}, \delta\beta)$ is observable if and only if the stacked matrix $[\mathbf{G} \ \Psi]$ has full column rank; equivalently, if and only if all three of the following hold: (i) $\ker \mathbf{G} = \{\mathbf{0}\}$ (the map gradient excites every position direction over the window); (ii) $\ker \Psi = \{\mathbf{0}\}$ (the compensation basis has full column rank); and (iii) $\text{range}(\mathbf{G}) \cap \text{range}(\Psi) = \{\mathbf{0}\}$ (no shared direction). Condition (iii) alone governs the position–compensation confounding: when it fails, a position shift with $\mathbf{G}\delta\mathbf{p} \neq \mathbf{0}$ is reproduced exactly by a compensation change.

Proof:

Two joint states $(\delta\mathbf{p}, \delta\beta)$ and $(\delta\mathbf{p}', \delta\beta')$ produce identical noise-free predictions iff $\mathbf{G}(\delta\mathbf{p} - \delta\mathbf{p}') = \Psi(\delta\beta' - \delta\beta)$, i.e. iff $[\mathbf{G} \ \Psi][(\delta\mathbf{p} - \delta\mathbf{p}')^\top \ (\delta\beta - \delta\beta')^\top]^\top = \mathbf{0}$. The unobservable subspace is therefore exactly $\ker[\mathbf{G} \ \Psi]$, and the joint state is observable iff $[\mathbf{G} \ \Psi]$ has full column rank; the first statement follows by taking $\delta\beta'$ free and $\delta\mathbf{p}' = \mathbf{0}$. For the three-condition equivalence: if (i) fails, take $\delta\mathbf{p} \in \ker \mathbf{G} \setminus \{\mathbf{0}\}$, giving the kernel element $(\delta\mathbf{p}, \mathbf{0})$; if (ii) fails, take $\delta\beta \in \ker \Psi \setminus \{\mathbf{0}\}$, giving $(\mathbf{0}, \delta\beta)$; if (iii) fails, take $\mathbf{G}\delta\mathbf{p} = \Psi\delta\beta \neq \mathbf{0}$, giving $(\delta\mathbf{p}, -\delta\beta)$. Conversely, let $(\delta\mathbf{p}, \delta\beta) \neq (\mathbf{0}, \mathbf{0})$ lie in the kernel, so $\mathbf{G}\delta\mathbf{p} = -\Psi\delta\beta$. If $\mathbf{G}\delta\mathbf{p} = \mathbf{0}$ then either $\delta\mathbf{p} \neq \mathbf{0}$ violates (i) or $\delta\beta \neq \mathbf{0}$ violates (ii); if $\mathbf{G}\delta\mathbf{p} \neq \mathbf{0}$ it is a nonzero shared direction, violating (iii). ■

Corollary 1 (Exogeneity and separability):

Suppose $\mathbf{A}_t$ is a function of aircraft attitude/fluxgate alone, exogenous to position. Then no column of Ψ is generated by the map gradient along the trajectory, and condition (iii) holds for generic trajectories and maps (in particular, it requires the window to satisfy $N \geq \dim \text{range}(\mathbf{G}) + \dim \text{range}(\Psi)$ and no Ψ column to be a linear combination of the map-gradient time series). Position–compensation separability is then independent of the basis dimension: enlarging an exogenous basis can slow the coefficient estimate through (ii) but cannot mask position. If instead a compensation feature contains the map value $h(\mathbf{p})$ (for example the scalar magnetometer itself), its column lies in $\text{range}(\mathbf{G})$ by construction, forcing the intersection nonzero and collapsing observability.

Remark 2 (TL rank deficiency as gauge freedom):

Condition (ii) fails for the classical TL basis itself: with unit direction cosines, the three induced diagonal columns sum to $B_t(u^2 + v^2 + w^2) = B_t$, collinear with the constant-bias column, the long-known collinearity of TL regression [23], [25]. This is a gauge freedom of the compensation, not a navigation defect: any $\delta\beta \in \ker \Psi$ leaves $c_t = \mathbf{A}_t^\top \beta_t$, and hence every prediction and the position estimate, unchanged. The prior $\mathbf{P}_0$ and the random-walk factors select one representative of each

gauge orbit. Separability of position from compensation is governed by (iii) alone and is unaffected.

Proposition 1 is exact but non-constructive: it does not yield a reliable scalar screening statistic, and in our experiments correlation-based proxies for the intersection (in-sample basis R^2 , out-of-sample prediction, and gradient-alignment indices) did not generalize across estimators or lines. The actionable consequence is Corollary 1: drive the compensation from exogenous features, which every estimator in Section VI does.

Remark 3:

The condition is geometric, not statistical: it depends on the ranges of $\mathbf{G}$ and Ψ over the window, hence on the flight path and the local map gradient, not on noise levels. It explains two failure modes seen in practice. First, over a locally flat map $\mathbf{G} \approx \mathbf{0}$ and condition (i) fails: position is unobservable regardless of the compensation, a map-information floor that bounds every estimator, and that we observe on the shortest line in Section VI. Second, an endogenous feature such as the scalar magnetometer reading itself makes a Ψ column collinear with $\mathbf{G}$, so the compensator silently absorbs the map signal; excluding such features (Section V) is necessary for the baseline to converge.

V. Reimplemented Online EKF+TL+NN Baseline

To compare against [12] on equal footing we reimplemented, rather than merely cite, the online EKF+TL+NN cold start. The filter augments the Pinson error state with the online TL coefficients β and the weights of a small feed-forward network, all propagated as EKF states and updated online against (3). Because neither the network architecture nor the natural-gradient training design of [12] is released, ours is a reimplementation of the published design principles, not a line-for-line replication. Naively initialized, the reimplementation diverged; three fixes, each addressing a distinct and independently diagnosed failure, were required.

(i) Bias-free network with DC initialization. The map returns the anomaly only, whereas the raw scalar reading includes a large constant offset. Left in the regression target, this offset drove the network output bias to $\sim 5 \times 10^4$ nT and the filter to NaN. We remove the mean by a short initial window, initializing the network output bias to the median measured-minus-predicted offset and scaling the target by its standard deviation.

(ii) Exogenous network inputs only. Using the scalar magnetometer as a network input creates exactly the endogenous feature of the Remark above: the network learns to reproduce the map and horizontal position collapses (we observed > 5 km error with a small measurement residual). Restricting the inputs to the attitude-derived TL basis restores observability, in direct agreement with Corollary 1.

(iii) De-trusting the map through the transient. During the cold-start transient the compensation is grossly wrong, so the map residual is unreliable; we inflate the measurement variance and the coefficient process noise for the warm-up, then relax them.

Concretely, the reimplemented filter augments the navigation state with the permanent TL group and a single-hidden-layer network of eight units acting on that group, giving 41 network-weight states carried in the EKF; the FOGM disturbance state uses a (σ, τ) of $(3 \text{ nT}, 180 \text{ s})$ and the measurement variance is 12^2 nT^2 after warm-up. The DC-removal step measures an interference offset of, e.g., -1724 nT on Mag 4 and $+310 \text{ nT}$ on Mag 5, which if left in the target is what drove the untreated filter to divergence. With these three fixes the filter reaches 40.0/17.5 m on Mag 4/5 (Table III): on Mag 4 this lies inside the 37 m to 58 m band spanned by the reported TL-only and TL+NN results, while on Mag 5 it is 2.5 m to 3.5 m above the reported values (14 m to 15 m), a functioning, close, but not exact reimplementation. Its late-half residual falls to 117 nT to 231 nT, and we use it as the baseline. We stress that the network only compensates; the fixes are what make it observable jointly with position, which is the same requirement Corollary 1 places on our own compensation features.

VI. Experiments and Results

A. Dataset, maps, and protocol

All experiments use the public SGL 2020 challenge dataset [9]: a Cessna 208B instrumented with a flight-grade INS, several onboard scalar magnetometers (a compensated tail-stinger sensor and uncompensated cabin sensors Mag 4 and Mag 5, all optically-pumped cesium-vapor units), a three-axis fluxgate, and reference truth. The flights serve distinct purposes, and we choose lines accordingly. Flt1007 is the free-flight mission, in which the aircraft flies freely within the mapped Renfrew (≈ 400 m) and Eastern/Perth (≈ 800 m) areas, so its lines are the closest to a navigation task and are what prior work [9], [4], [12] reports; 1007.06 is our primary line for that reason. Flt1003 is a crustal-field measurement flight (survey traverses at 400 and 800 m); its lines 1003.02 and 1003.08 are straighter than the free-flight legs. Flt1006 is a calibration-maneuver flight, so we set its single short line aside (Section VI-D), and the map-building flights (Flt1004–1005) are not used. Two honest points remain. First, none of these is a dedicated long, straight-and-level operational transit; the free-flight and survey legs carry more attitude variation, which stresses compensation but also aids TL observability by exciting the regressor (Section IV), so performance on a pure operational cruise is not directly established here and is future work. Second, the lines span two altitude regimes: four at the ≈ 400 m survey-drape altitude and 1007.02 at ≈ 800 m, where upward continuation attenuates the anomaly. Map matching uses the Eastern (“E”) and Renfrew (“R”)

TABLE I

Estimator settings used for all cold-start results (values from the released code). The same $\mathbf{P}_0$, $\mathbf{Q}$, R , and TL settings drive both the causal EKF and the window smoother.

Group	Parameter	Value
Measurement	noise std. $\sqrt{R}$	12 nT
FOGM S	(σ_S, τ)	3 nT, 180 s
Initial $\mathbf{P}_0$	pos. / alt. std.	0.1 / 1.0 m
	velocity std. (per axis)	1.0 m s ⁻¹
TL prior / walk	$\mathbf{P}_0^\beta$; $\mathbf{Q}^\beta$	$\mathbf{I}(1.0)^2$; $\mathbf{I}(1.0)^2 \Delta t$
TL basis	terms (App. B)	P+I+E+bias (19)
Window	L_w / L_o / Δt [s]	300 / 90 / 0.1
Robust	Huber δ	1.345
	IRLS	2nd iter. on, $w \in [10^{-6}, 1]$
Gauss–Newton	max. iter. / tol.	5 / 10^{-4}

high-resolution anomaly grids, upward-continued to flight altitude with the IGRF core field restored. The metric is horizontal distance-RMS (DRMS) of position error after a 10 min warm-up. Cold start means the TL coefficients are initialized to zero with no calibration flight. For every comparison the EKF and FGO estimators use identical dynamics, map interpolation, and noise settings (Table I); the window smoother additionally applies the Huber kernel (10), whose isolated contribution is reported separately in Table III (37.0→32.6 m on Mag 4), so the reader can attribute the remaining margin to the estimator structure. Unless noted, the window smoother uses $L_w = 5$ min and $L_o = 90$ s; L_w was selected on line 1007.06 (Fig. 9) and then held fixed for every other line, so the breadth results of Table IV are not per-line tuned. All numbers are produced by continuous-integration scripts on the code branch and are reproducible from it.

B. Batch estimator on real data

We first validate the estimator on survey line 1003.02, from the crustal-field measurement flight Flt1003 (63 min), with the compensated stinger magnetometer. This is a deliberately controlled check rather than the navigation demonstration: because the interference is already removed, the comparison isolates the estimator and solver from the cold-start compensation problem; the navigation showcase proper is the free-flight line 1007.06 of Section VI-C. The batch factor graph improves horizontal DRMS from the EKF’s 28.3 m to 14.0 m (Huber kernel), against 114.6 m free-inertial, a 51 % reduction over the EKF. Fig. 6 shows the anomaly map and the flight line, with a zoom on the fine anomaly structure the map-matching factor exploits; the estimator comparison itself is carried by the error and CDF plots below rather than the map, since a metre-level track error is not resolvable at map scale. Fig. 7 plots the horizontal error over the line: the FGO estimate (blue) stays below the EKF (red) for essentially the entire flight, the shaded region marking the improvement. Fig. 8 gives the empirical error distribution: the FGO curve reaches its 95th percentile at less than half

TABLE II

Batch estimator on survey line 1003.02 (Flt1003 measurement flight, compensated stinger), a controlled estimator/solver check, not the navigation result; horizontal DRMS [m]. Free-inertial reference 114.6 m. The RTS and GN/QR solvers agree, confirming a common MAP estimate.

Segment	Estimator	DRMS
full line	EKF	28.3
full line	FGO (RTS)	14.3
full line	FGO (RTS, Huber)	14.0
10 min	EKF	47.9
10 min	FGO (RTS)	18.4
10 min	FGO (GN/QR)	19.4
10 min	FGO (GN/QR, Huber)	18.4

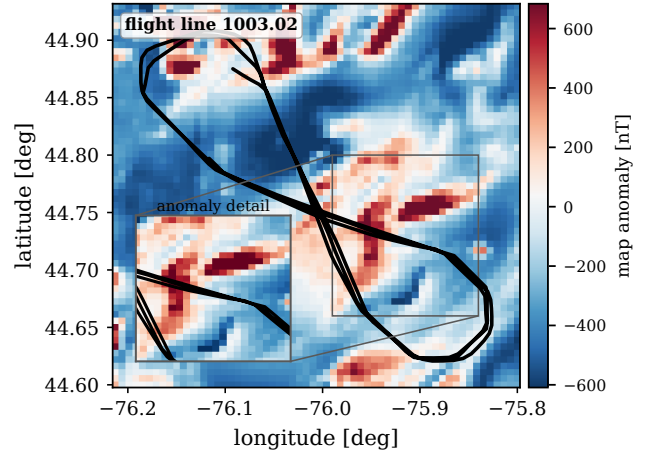


Fig. 6. Eastern anomaly map and survey line 1003.02 (Flt1003 measurement flight), providing geographic context; the inset zooms the fine anomaly structure the map-matching factor exploits along the line. The estimator comparison is deferred to Figs. 7–8: at the survey ground speed a metre-level track error is smaller than one 12 s sample and cannot be resolved on a map.

the error of the EKF, and both bound the free-inertial error by an order of magnitude. On a 10 min segment the iterated-RTS and sparse-GN/QR solvers agree to within 1 m (Table II), confirming that both numerically reach the same MAP estimate as predicted by Lemma 1. The Huber kernel changes the batch result by less than 0.3 m here, as expected for the already-compensated stinger sensor whose residuals are close to Gaussian; its value appears in the cold-start cases below, where transient outliers are large.

C. Cold-start line vs. online EKF+TL+NN

The central comparison is the cold start on uncompensated cabin magnetometers, where the interference is present and must be learned online. Table III and Fig. 10 report the primary evaluation line 1007.06 (87 min, full length) on Mag 4 and Mag 5. A static-TL batch forces a single coefficient vector over the whole line; it underfits the time-varying interference and degrades to 123.7/68.1 m. Allowing the compensation to adapt with a

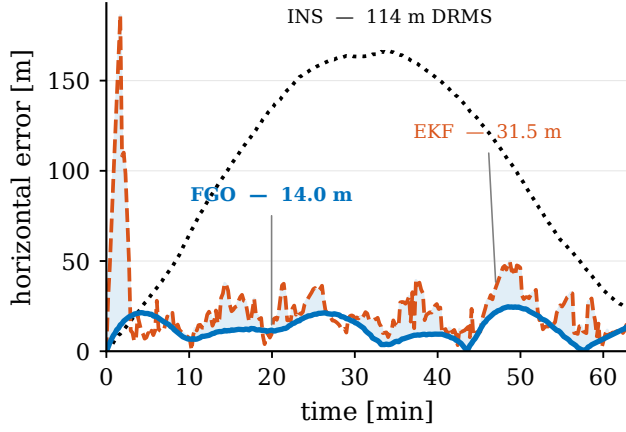


Fig. 7. Horizontal position error on survey line 1003.02 (Flt1003). The batch factor graph (FGO, blue) holds a tighter bound than the EKF (red) across the line; the shaded band marks the FGO improvement, and free inertial (INS) diverges. The labelled DRMS values are computed over the full line, without the warm-up exclusion of Table II.

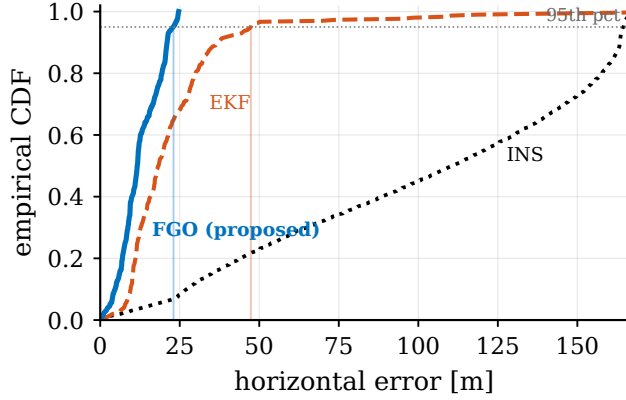


Fig. 8. Empirical CDF of horizontal error on survey line 1003.02 (Flt1003; full line, as in Fig. 7). The proposed FGO reaches its 95th percentile at less than half the EKF's error; both bound the free-inertial error by an order of magnitude.

5 min window recovers it to 37.0/15.1 m, and the Huber kernel further trims transient outliers to 32.6/14.2 m. Our reimplemented online EKF+TL+NN reaches 40.0/17.5 m (Section V details how close this is to the reported operating point), while the reported TL-only EKF is weaker still on Mag 4. The window smoother beats the reimplemented NN-augmented filter on both sensors without any neural network, using only the linear TL basis carried as graph variables; against the values reported in [12] it is 4.4 m better on Mag 4 and 0.2 m behind on Mag 5. This comparison against the published values is on the single line 1007.06, the only line [12] reports; our reimplemented EKF+TL+NN carries the baseline to the other lines in the breadth study (Table IV). We attribute the margin to the window look-ahead: the coefficients are identified from data on both sides of each committed epoch, whereas the EKF must commit with past data only.

The window length itself trades under-determination against staleness. Fig. 9 plots DRMS at the three lengths evaluated on this line: the 2 min window under-determines

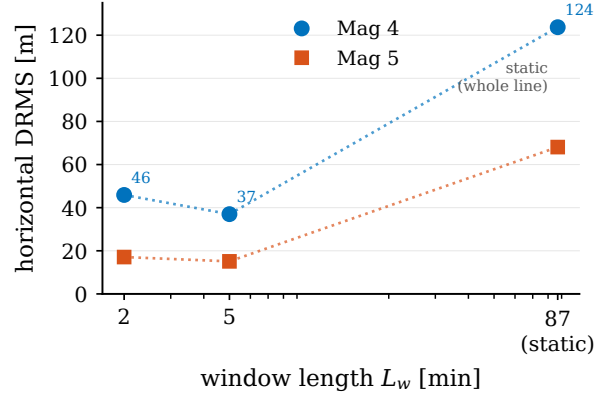


Fig. 9. Window length versus accuracy on the free-flight navigation line 1007.06 (Flt1007, cold start), at the three evaluated lengths. The 5 min window balances the under-determined 2 min window against the drift-blind static batch (full line).

TABLE III

Cold start on the free-flight navigation line 1007.06 (Flt1007), uncompensated cabin magnetometers; DRMS [m]. This is the primary navigation result. INS 318 m; EKF on compensated Mag 1 $\approx$ 20 m.

Method (no NN unless noted)	Mag 4	Mag 5
FGO-online, static batch TL	123.7	68.1
FGO-online, window 2 min	45.9	17.1
FGO-online, window 5 min	37.0	15.1
FGO-online, window 5 min + Huber	32.6	14.2
EKF+TL+NN, reimplemented [12]	40.0	17.5
EKF+TL+NN, reported [12]	37	14
EKF+TL-only, reported [12]	58	15

the coefficients, the full-line static batch cannot track their drift, and the 5 min window is the best of the three. The tradeoff mirrors the expressiveness discussion of Section IV and makes L_w a physically interpretable knob rather than a free hyperparameter, though a denser sweep (and other lines, where the balance can shift toward longer windows) is left for the revision.

These cold-start numbers also settle a question the formulation raises: with no neural network, how does the estimator cope with the large fraction of the cabin interference that the linear TL basis cannot span. The mechanism is the one built in Section III-A, where the time-varying coefficients, the FOGM disturbance state, and the robust kernel together absorb that remainder as a structured robust regression. The trade it makes is visible in the numbers. The network buys a lower compensation residual, while the window, the disturbance state, and the robust kernel together buy navigation robustness at a higher residual. This is why the uncompensated Mag 4 DRMS of tens to hundreds of metres sits well above survey-grade compensation and still bounds the inertial drift. A learned residual could be added later as an extra factor on the same measurement, for example a Gaussian-process or small-network term, which would place the network's expressiveness inside the graph rather than inside a filter; we leave that to future work.

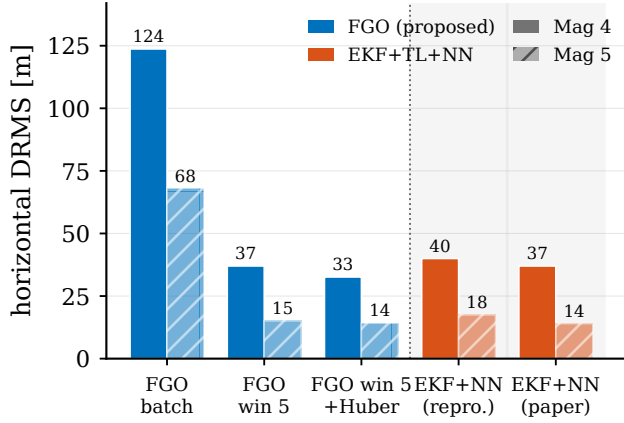


Fig. 10. Free-flight navigation line 1007.06 (Flt1007), cold start on the uncompensated cabin magnetometers. The NN-free fixed-lag window (left of the divider) compares against the reimplemented and reported EKF+TL+NN (shaded, right).

D. Breadth across lines, flights, and maps

Table IV runs the same cold-start TL model on four long lines spanning two flights and two maps, free-flight navigation legs (Flt1007) and survey traverses (Flt1003), grouped by purpose, against two causal baselines: the plain online-TL EKF, and the reimplemented EKF+TL+NN of Section V, the strong causal method that a fair comparison demands.

The two baselines behave very differently, and the contrast is the point. The plain online-TL EKF is fragile: on the noisy Mag 4 it diverges from the cold start to tens of kilometres on two lines and runs off the map on a third. The neural network fixes exactly this: the EKF+TL+NN stays finite on all eight cases, confirming that its residual network is what buys cold-start robustness for the causal filter. The window smoother matches that robustness without any neural network (it is bounded on all eight) and improves on accuracy: across the four full-length lines (63 min to 87 min, eight mag-cases) it is the best of the three on all eight, beating the NN-augmented filter by 4 m to 76 m (Fig. 11).

We also ran the short calibration-maneuver line 1006.08 (Flt1006) but set it aside rather than tabulate it: it is only 14 min long, so the 10 min warm-up leaves a 4 min window over which none of the three DRMS values is reliable, and its low map-information floor (the Remark in Section IV) makes every method weak. It reproduces the same causal-EKF divergence, but we draw no accuracy conclusion from it: in particular the NN’s 34.6 m on its Mag 5, against 117 m to 122 m for the TL-only methods, is not evidence of an NN accuracy advantage on a window this short.

Two effects in the reliable cases are worth naming. First, the EKF+TL+NN follows the reproduced design in feeding the network only the permanent TL group, whereas the online-TL EKF and the window smoother both carry the full permanent-induced-eddy-bias basis; on the well-conditioned Mag 5 those linear higher-order

TABLE IV

Breadth: cold-start cabin magnetometers, DRMS [m], on the SGL 2020 navigation-usable lines. “Purpose”: FF = free-flight (Flt1007), SV = survey traverse (Flt1003). “div.” = divergence > 10 km; “err.” = run left the map. Bold = best of the three baselines. The short calibration-maneuver line 1006.08 (Flt1006) is set aside and not tabulated (see text).

Line	Purp.	Map	alt [km]	Mag	EKF online	EKF+ TL+NN	FGO win
1007.06	FF	R	0.4	4	46.7	42.2	32.7
1007.06	FF	R	0.4	5	17.8	17.8	13.8
1007.02	FF	E	0.8	4	div.	114.8	38.6
1007.02	FF	E	0.8	5	31.6	29.3	14.5
1003.02	SV	E	0.4	4	div.	99.1	42.6
1003.02	SV	E	0.4	5	28.1	33.8	21.7
1003.08	SV	R	0.4	4	err.	46.4	26.1
1003.08	SV	R	0.4	5	21.1	18.8	12.4
FGO best (all counted lines)							8 / 8

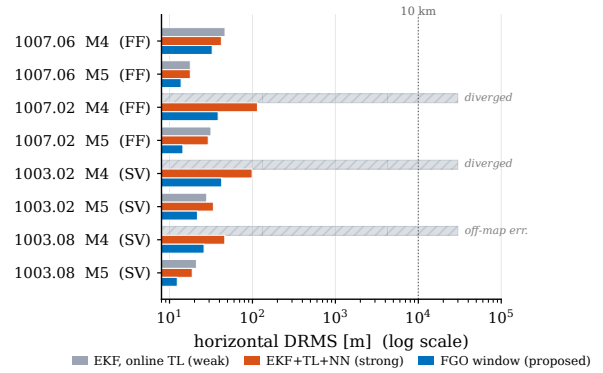


Fig. 11. Breadth study (Table IV) on a log DRMS axis: weak online-TL EKF (gray, hatched where it diverges beyond 10 km), strong EKF+TL+NN (red), and the NN-free FGO window (blue). The NN keeps the causal filter bounded; the window matches that and is best of the three on all 8 full-length cases. The short calibration line 1006.08 is set aside and not shown (see text).

terms matter, so the NN filter can fall behind even the plain EKF (1003.02: 33.8 vs 28.1 m) when its network has not yet learned what the dropped linear terms would have supplied directly. Second, and relatedly, the network’s extra online-learned states add estimation variance where the linear basis already suffices, a cost the NN-free window avoids. Both effects favor a linear compensation carried in a smoother over a learned one carried in a filter, on this benchmark. One caveat on provenance: the EKF+TL+NN is our reimplement, matching the reported 1007.06 operating point to within a few metres (Section V), but the original’s per-line numbers on the other lines are unpublished, so the comparison is against a faithful, not the authors’, filter.

Two conclusions follow, and they are worth separating because a single-line comparison conflates them. The first is about robustness. Keeping a causal filter bounded at cold start is not unique to the window, since the neural network does it too, so the blanket claim that causal filters diverge is false for the state-of-the-art filter. What

TABLE V

Estimator families for cold-start map-matching compensation. ✓ = yes, ✗ = no, (∼) = limited.

Estimator	Future data	Comp. state	Non-Gauss.	Bounded latency
Causal EKF [4]	✗	(∼)	✗	✓
Particle / grid filter [7]	✗	✗	✓	✓
Online EKF+TL+NN [12]	✗	✓	(∼)	✓
Batch FGO	✓	✓	✓	✗
Proposed (fixed-lag FGO)	✓	✓	✓	✓

the window adds is that it reaches the same bounded regime without the network, its online-training machinery, or the verification burden that comes with it. The second is about accuracy, where the window is the best of the three on all eight full-length cases. This margin is real, but it should be read with care, because our EKF+TL+NN is a reimplement and part of its deficit is that its online-learned states add variance on the well-conditioned sensor, where the linear TL basis already suffices. The defensible reading is that a linear estimator with no network, given future context through the window, is enough both to survive the cold start and to match or beat an NN-augmented causal filter on this benchmark. Table V places the estimator families along the axes that matter for cold-start compensation.

The observability condition of Section IV accounts for the pattern. Exogeneity (Corollary 1) is why the scalar magnetometer had to be excluded from the network inputs for the baseline to converge, and why the attitude-only TL basis stays safe at any dimension. The map-information floor of the Remark is why the short calibration line 1006.08 is hard for every method, since there is little independent map gradient to exploit over its brief window, and it is the reason we set that line aside. The condition is geometric and predictive only through these qualitative rules; a scalar screening statistic we tried did not generalize, which we report as a negative result.

E. Sensor-error factors (simulation)

To exercise the sensor-error factors of Section III we inject the optically-pumped-magnetometer error model of [13], [14] into a simulated flight (free-inertial reference 31.1 m): a cosine-harmonic heading error at orders $\{1, 2, 4\}$ (7.6 nT RMS), a fluxgate hard-iron bias, a linear drift, and the equatorial-dead-zone noise amplification $1/\max(|\cos \psi|, \epsilon)$. We then enable the corresponding graph variables cumulatively; Table VI reports the result. With no sensor-error states the estimate is worse than free inertial (42.9 m): the unmodeled error corrupts the map aiding, exactly the mechanism of Proposition 1. Enabling all four factors reduces the error to 6.2 m, an 86 % reduction, monotone in this order. Because the factors are enabled cumulatively in a single order, Table VI attributes marginal gains only for that order: the largest single-step drop comes from the heading harmonics (42.9 → 18.9 m),

TABLE VI

Sensor-error factors (simulation), enabled cumulatively; horizontal DRMS [m]. Free-inertial reference 31.1 m. Values for the non-robust solver; the Huber solver agrees within 2 m.

Graph variables enabled	DRMS
none (map aiding only)	42.9
+ heading harmonics $\{1, 2, 4\}$	18.9
+ dead-zone weight	15.5
+ hard-iron bias	8.1
+ linear drift (full)	6.2

followed by the dead-zone weighting (→ 15.5 m, now load-bearing because the injected dead-zone noise is heteroscedastic), the hard-iron bias (→ 8.1 m), and the linear drift, whose rate the estimator recovers cleanly ($\hat{\gamma}_1 = 0.018$ versus a true 0.020). The heading coefficients themselves are poorly identified ($\hat{c}_1 = 3.8$ versus a true 10): level flight sweeps the sensor-field angle by only about 12° , so the harmonic regressor is barely excited, an observability property of the trajectory, consistent with the Remark of Section IV, that a maneuvering flight would relieve, and the gain to navigation comes from the fitted combination over the observed angle range, not from the individual coefficients. The Huber kernel gives similar values (within 2 m; it helps most at the dead-zone step). Two caveats bound this study: it is simulation-only, and the injected errors share the estimator’s functional form, so it demonstrates that the graph formulation can carry and recover such states, not that these models capture real sensor behavior.

F. Consistency and statistical significance

Each real line provides a single physical realization, so its DRMS is a point estimate and its measurement noise cannot be resampled. To assess statistical significance and, more importantly, whether the reported covariance is credible, we run a Monte-Carlo study in simulation, where repeated realizations are valid. We fix one simulated trajectory over the high-resolution Eastern anomaly map and, for each of 30 seeds, draw an independent INS-error realization and an independent clean scalar-measurement realization, then run the causal EKF and the batch factor graph on the same draw. The clean compensated measurement isolates the navigation covariance from the cold-start compensation transient.

Averaged over the 30 seeds, after a 30 s warm-up, the horizontal DRMS is 7.4 m for the factor graph and 9.3 m for the EKF, with 95 % confidence half-widths of 1.6 and 1.3 m. On this clean sensor both estimators sit near the map-limited floor, so the intervals overlap and the gap is modest; the large accuracy separation is the cold-start regime of Sections VI-C and VI-D, not this one. The purpose of this study is consistency.

For consistency we use the per-axis two-degree-of-freedom position NEES, $(e_N/\sigma_N)^2 + (e_E/\sigma_E)^2$, whose expectation under a consistent covariance is 2. Averaged

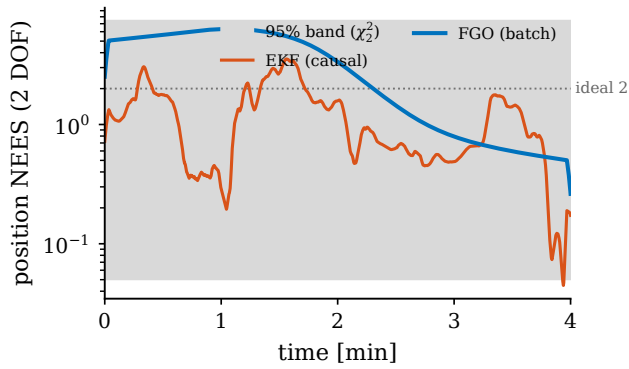


Fig. 12. Monte-Carlo consistency. Per-axis two-degree-of-freedom horizontal-position NEES against time for a representative simulated run, for the causal EKF and the batch factor graph, with the 95 % χ^2_2 band shaded and the ideal value 2 dotted. Both estimators stay inside the band, so neither reported covariance is over-confident.

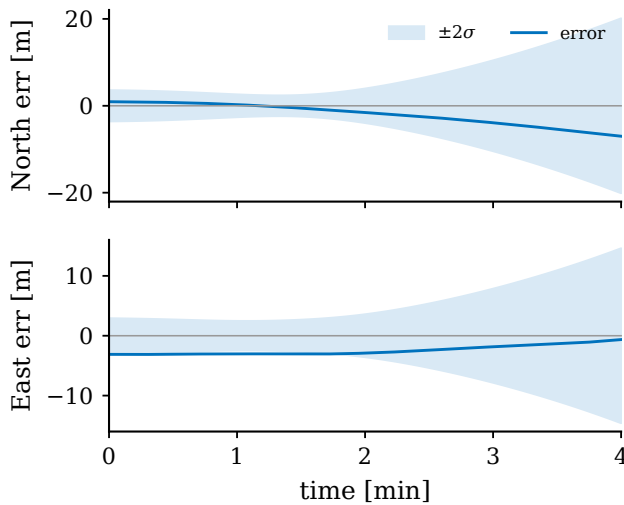


Fig. 13. Covariance envelope. North and East position error for a representative simulated factor-graph run inside the estimator $\pm 2\sigma$ bands. The error stays within the reported uncertainty throughout, consistent with the conservative ANEES of 1.5.

over all seeds and epochs the ANEES is 1.54 for the factor graph and 1.11 for the EKF, both below the ideal 2. A value below 2 means the reported uncertainty is conservative and bounds the true error with margin, whereas a value above 2 would signal an over-confident covariance. Neither estimator is over-confident, and the smoother covariance is the better calibrated of the two, closer to the ideal. Fig. 12 shows the NEES staying inside the 95 % χ^2_2 band throughout a representative run, and Fig. 13 shows the North and East position error contained within the estimator $\pm 2\sigma$ envelope. Two caveats bound the reading. The NEES uses the per-axis (diagonal) covariance, since the cross term is not exposed by the evaluated filter output, and the study is a matched-model simulation, so it certifies internal consistency rather than robustness to model mismatch. The conservative bias indicates that the process noise could be tightened, which is the safe direction in which to err.

G. Computational considerations

Each window is a sparse nonlinear least-squares problem of fixed size $O(L_w(n + m))$; the block-bidiagonal Jacobian of the two chains (Fig. 3) is factored in $O(L_w)$ by the square-root solver [32], and the number of Gauss–Newton iterations is small (typically 3–5) because the problem is mildly nonlinear away from observability collapse. The per-window cost is therefore constant in the line length and the overall complexity is linear in the number of epochs, matching a filter’s asymptotic cost while retaining the look-ahead. Memory is bounded by the window, not the flight. The one-time overhead relative to an EKF is the window factorization and the IRLS reweighting, which in our unoptimized research implementation runs comfortably faster than real time on the SGL sampling rate.

These same properties make the estimator compatible with real-time operation. Its committed output lags real time by the overlap L_o , here 90 s, a latency that suits the map-aiding role, in which the INS supplies the high-rate solution and the smoother corrects it periodically, and that the designer trades against accuracy through L_o . The factor interface is modular, so the compensation, sensor-error, and map factors can be added or removed without touching the solver, and one implementation then covers the calibrated case with frozen coefficients, the cold-start case with adaptive coefficients, and the augmented-sensor case. Integrating it with an operational system amounts to feeding the INS mechanization, fluxgate, and scalar magnetometer streams into the graph and reading back the corrected position at the commit rate. The main tuning burden is the two window parameters and the compensation process noise $\mathbf{Q}^\beta$, all of which have clear physical meaning.

VII. Conclusion

Casting online Tolles–Lawson aeromagnetic compensation as time-varying factor-graph variables in a fixed-lag sliding-window smoother yields a cold-start calibrator that uses no neural network. On real SGL 2020 data it matches or beats a reimplemented online EKF+TL+NN on the primary free-flight line and generalizes across four lines, two flights, and two maps, staying bounded where the causal EKF diverges. An exact observability condition and its exogeneity corollary explain this behaviour and guide the choice of compensation features.

Several limitations bound the study and set the agenda for further work. Each real line is a single physical realization, so its DRMS is a point estimate; a simulation Monte-Carlo (Section VI-F) establishes covariance consistency and statistical significance, but repeating the real lines and extending to the full SGL line set remain out of reach with a single recorded flight per line. The comparison covers a causal online-TL EKF and a reimplemented online EKF+TL+NN, but not a like-for-like grid or particle-filter map-matching baseline [16], [7], which would sharpen the

claim about estimator structure. The sensor-error factors are validated in simulation only, and a flight demonstration with genuine optically-pumped-magnetometer heading error is still needed. Most important, although the primary line comes from the free-flight mission and is the community's navigation benchmark, none of the SGL 2020 lines is a dedicated long, straight-and-level operational transit; their attitude content both stresses compensation and, by exciting the TL regressor, aids its observability, so performance on a pure operational cruise with little attitude variation is not established here and is the natural next step. Short lines remain hard for every method, which is an information limit rather than an estimator one. None of these qualifications affects the central and reproducible finding, that a linear smoother with no neural network is the best of three cold-start estimators on all eight full-length cases, bounded where the plain causal filter diverges and more accurate than an NN-augmented causal filter that does use one.

Appendix A Pinson Error-State Dynamics

The navigation-error block of Φ_t in (2) follows the Pinson model in a local north-east-down frame [2], [1]. With position, velocity, and attitude errors ($\delta\mathbf{p}, \delta\mathbf{v}, \psi$) and sensor biases ($\mathbf{b}_a, \mathbf{b}_g$), the continuous error dynamics are

$$\begin{aligned}\dot{\delta\mathbf{p}} &= \mathbf{F}_{pp} \delta\mathbf{p} + \delta\mathbf{v}, \\ \dot{\delta\mathbf{v}} &= \mathbf{F}_{vp} \delta\mathbf{p} + \mathbf{F}_{vv} \delta\mathbf{v} - [\mathbf{f}^n \times] \psi + \mathbf{C}_b^n \mathbf{b}_a, \\ \dot{\psi} &= \mathbf{F}_{\psi p} \delta\mathbf{p} + \mathbf{F}_{\psi v} \delta\mathbf{v} - [\boldsymbol{\omega}_{in}^n \times] \psi - \mathbf{C}_b^n \mathbf{b}_g,\end{aligned}\quad (12)$$

where $\mathbf{f}^n$ is the specific force, $\mathbf{C}_b^n$ the body-to-nav rotation, $\boldsymbol{\omega}_{in}^n$ the inertial rate, $[\cdot \times]$ the skew operator, and the $\mathbf{F}_{\bullet\bullet}$ blocks collect the transport-rate, gravity-gradient, and Coriolis terms. The biases follow first-order Gauss–Markov or random-walk models, as does the scalar magnetic disturbance state S with correlation time τ . The discrete $\Phi_t = \exp(\mathbf{F}_t \Delta t)$ and $\mathbf{Q}_t$ are obtained by standard van Loan discretization [42]. Only $\mathbf{g}_t^\top$, the position row of the measurement Jacobian (9), couples this block to the map.

Appendix B Tolles–Lawson Basis

The basis row $\mathbf{A}_t$ of (6) is built from the fluxgate direction cosines $\hat{\mathbf{b}} = [u \ v \ w]^\top$ (with $u^2 + v^2 + w^2 = 1$) and the scalar field magnitude B [8], [23]. The permanent moment contributes the three cosines (u, v, w); the induced magnetization, symmetric in the field, contributes the six products $B(u^2, uv, uw, v^2, vw, w^2)$; and the eddy-current field, proportional to the rate of change of the cosines, contributes the nine products $B(u\dot{u}, u\dot{v}, \dots, w\dot{w})$. Stacking these gives an eighteen-term basis, optionally augmented by a constant bias, and $c_t = \mathbf{A}_t^\top \boldsymbol{\beta}_t$ is linear in the coefficients $\boldsymbol{\beta}_t$, the property that lets them enter the factor graph as ordinary linear-measurement variables. In the cold-start regime the calibration flight that classically

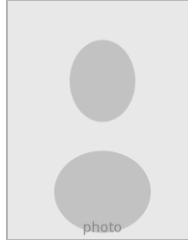
fixes $\boldsymbol{\beta}$ is unavailable, which is precisely why we carry it as a graph variable.

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The experiments use the publicly released SGL 2020 challenge dataset [9].



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